

TN POLICE HACKATHON 2023

SOCIALEYE

ABSTRACT

Team Name: T3tra

Team members:

- Varun, III B.Tech Computer Science and Engineering, KITS.
- Mukesh Kumar, III B.Tech Computer Science and Engineering, KITS.
- Bewin, II B.Tech Computer Science and Engineering, KITS.
- S Venkatesh, III B.Tech Computer Science and Engineering, KITS.

Problem Statement:

Social media video analysis

To classify the Metadata information available in the YouTube videos such as Genre, Language, People etc.

Objective:

To examine the video and extract the metadata

Solution :

We propose to develop a digital forensic analysis software to analyze a YouTube URL and extract usable information about the video and the uploader of the video. Our solution involves the following steps:

- **Data extraction:** Information for further investigation will be extracted from YouTube using automated (web scraping API) and proprietary data collection tools.
- **Information analysis:** The following data collected will analyzed during this step.

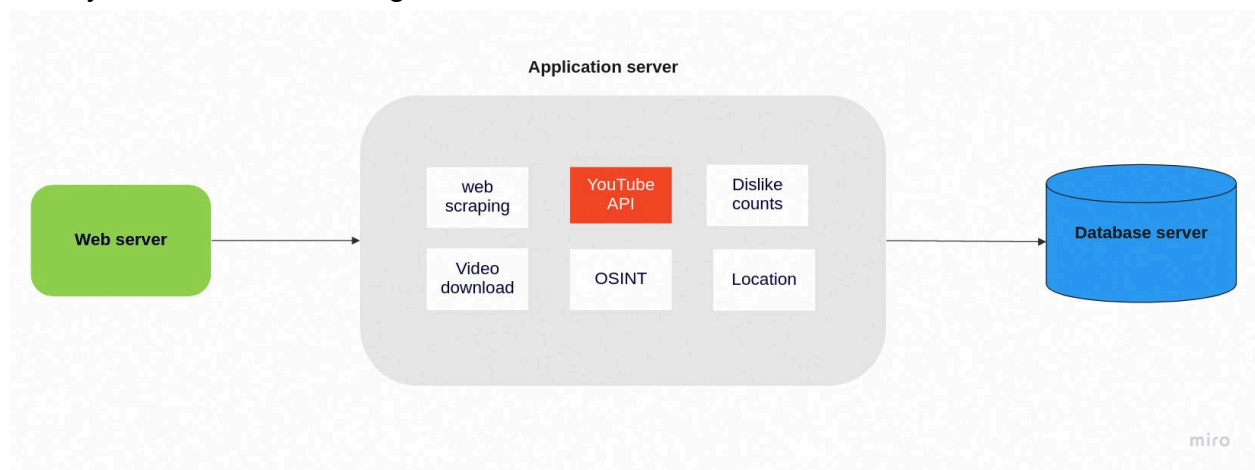
Invariant Data

- Email of uploader
- Phone Number of uploader
- Local Time Zone
- Last login time
- Subscriber
- Dislikes
- Other Social Profiles

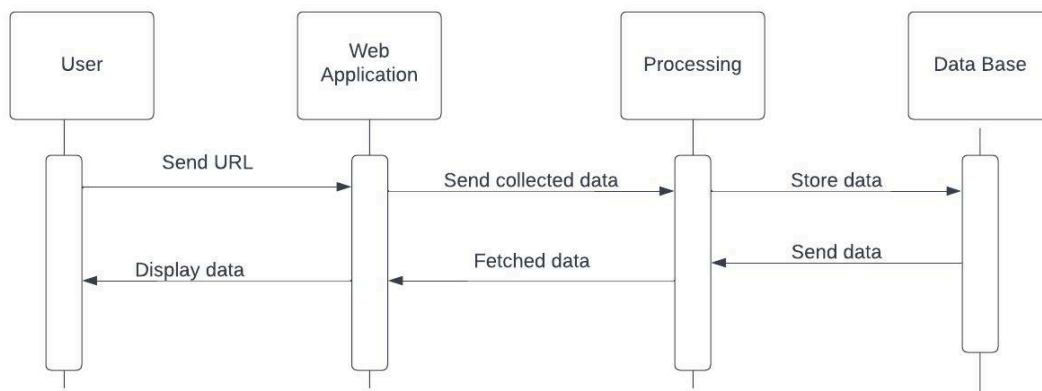
Dynamic Data

- Speech to text
- Sensitive information
- Spam
- Fake or legitimate Profile
- Fake video content
- Fake thumbnail
- **Report Generation:** A comprehensive report with the consolidated information will be presented to the authorities for further processing.

The system architecture is given below.



The process flow diagram is given below.



Highlights:

- **YouTube Metadata** - analyzing the YouTube metadata
- **Keywords Searching** - Searching with keywords
- **Dislike Count** - Number of dislike counts which has been disabled by YouTube
- **Video Downloader** - Downloading the particular video
- **Profile OSINT** - Mapping a particular person using the result information

Requirements:

Software:

- React
- MongoDB
- Database server
- Osint Tools
- YouTube API

Hardware:

- Web server: Intel Xeon 3 GHz with 8 GB and 256 GB HDD
- Database server: Intel Xeon 3 GHz with 8 GB and 256 GB HDD

Roadmap :

- UI/UX Design: 10th March, 2023 to 12th March, 2023
- Code Development: 12th March, 2023 to 16th March, 2023
- Presentation: 17th March, 2023
- Hackathon: 28th and 29th March, 2023

Team Skill Set:

Mukesh Kumar

- Front-end Development
- Network Configuration

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Bewin

- Front-end Development
- Back-end Development

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Varun

- Machine Learning
- Programming

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S Venkatesh

- Server Configuration
- Networking

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